

Industry 5.0 Human-Machine Collaboration - Embedded C Simulation

```
#include <stdio.h>
#include <stdlib.h>

// Function Prototypes
void HumanMachineCollaboration(int human, int machine, int emergency);

int main()
{
    int humanPresence;
    int machineReady;
    int emergencyStop;
    char choice;

    printf("=====\n");
    printf(" INDUSTRY 5.0 HUMAN-MACHINE COLLABORATION\n");
    printf(" Embedded C Real-Time Simulation\n");
    printf("=====\n");

    do
    {
        printf("\nEnter Human Presence (1=Detected, 0=No Human): ");
        scanf("%d", &humanPresence);

        printf("Enter Machine Status (1=Ready, 0=Not Ready): ");
        scanf("%d", &machineReady);

        printf("Emergency Stop? (1=Pressed, 0=Normal): ");
        scanf("%d", &emergencyStop);

        HumanMachineCollaboration(humanPresence, machineReady, emergencyStop);

        printf("\nRun Again? (y/n): ");
        scanf(" %c", &choice);

    } while(choice=='y' || choice=='Y');

    printf("\nSimulation Ended.\n");
    return 0;
}

void HumanMachineCollaboration(int human, int machine, int emergency)
{
    int motor = 0;
    int safetyLED = 0;

    printf("\n-----\n");

    if(emergency){
        motor=0; safetyLED=1;
        printf("Emergency Stop Activated!\n");
        printf("Machine Stopped Immediately.\n");
    } else if(human && machine){
        motor=1; safetyLED=0;
        printf("Human Detected.\n");
        printf("Machine Ready.\n");
        printf("Collaborative Mode Enabled.\n");
        printf("Robot Assisting Human Safely.\n");
    } else if(human && !machine){
        motor=0; safetyLED=1;
        printf("Human Present.\n");
        printf("Machine Not Ready.\n");
        printf("Waiting for Machine Initialization.\n");
    } else if(!human && machine){
        motor=1; safetyLED=0;
        printf("No Human Present.\n");
        printf("Machine Running in Automatic Mode.\n");
    } else {
        motor=0; safetyLED=1;
        printf("System Idle.\n");
        printf("Waiting for Human or Machine.\n");
    }

    printf("\n----- OUTPUT STATUS ----- \n");
```

```

    printf("Motor          : %s\n", motor ? "ON":"OFF");
    printf("Safety Indicator: %s\n", safetyLED ? "ON":"OFF");
    printf("-----\n");
}

```

Sample Output Screenshot

The screenshot displays the Programiz C Online Compiler interface. On the left, a sidebar contains icons for various programming languages: C, C++, Java, JavaScript, Python, PHP, Ruby, Swift, Kotlin, and TypeScript. The main editor area shows a C program named 'main.c' with the following code:

```

73  motor = 0;
74  safetyLED = 1;
75
76  printf("Human Present.\n");
77  printf("Machine Not Ready.\n");
78  printf("Waiting for Machine Initialization.\n");
79  }
80  else if(!human && machine)
81  {
82      motor = 1;
83      safetyLED = 0;
84
85      printf("No Human Present.\n");
86      printf("Machine Running in Automatic Mode.\n");
87  }
88  else
89  {
90      motor = 0;
91      safetyLED = 1;
92
93      printf("System Idle.\n");
94      printf("Waiting for Human or Machine.\n");
95  }
96
97  printf("\n----- OUTPUT STATUS ----- \n");
98  printf("Motor          : %s\n", motor ? "ON" : "OFF");
99  printf("Safety Indicator: %s\n", safetyLED ? "ON" : "OFF");

```

On the right, the 'Output' panel shows the execution results:

```

=====
INDUSTRY 5.0 HUMAN-MACHINE COLLABORATION
Embedded C Real-Time Simulation
=====

Enter Human Presence (1=Detected, 0=No Human): 1
Enter Machine Status (1=Ready, 0=Not Ready): 1
Emergency Stop? (1=Pressed, 0=Normal): 0

-----
Human Detected.
Machine Ready.
Collaborative Mode Enabled.
Robot Assisting Human Safely.

----- OUTPUT STATUS -----
Motor          : ON
Safety Indicator: OFF

-----
Run Again? (y/n): |

```